2025/2026

# CUSTOMER CHURN PREDICTION IN THE TELECOM INDUSTRY

DATA MINING

Lina Brahmi

# TABLE OF CONTENTS


I Executive summary 3

II Problem Statement 3

III Dataset Description 4

VI Methodology 5

VI-1 Random Forest

VI-2 Data Processing

VI-2 Python Code

V Model Results and valuation 6

# I Executive Summary

Customer Churn is the phenomenon where subscribers cancel their plans. This poses an issue for telecom companies specifically as it is more costly to acquire a new customer than to retain an already existing one.
Thus, accurately predicting whch customer are more at risk of leaving the allows businesses to intrevene proactively

This project builds a full machine learning pipeline using random forest algorithm to predict customer churn from a telecom dataset of 7032 customers to identify the most influential risk drivers and segment customers to facilitate decision making.

## Results at a glance

| | |
|---|---|
| Dataset | 7032 telecom customers with 26.6% churn rate |
| Algorithm | Random Forest (100 trees, max depth = unlimited) |
| ROC score | ROC-AUC Score: 0.815 |
| Overall Accuracy | Accuracy: 78.54% |
| Top churn drivers | TotalCharges 0.198838 |

# II Problem Statement

A mid-sized telecom company observes that roughly 1 in 4 customers leaves each quarter. The business challenge is twofold:
•Identify customers who are likely to churn before they actually do.
•Understand which features drive churn so that targeted business actions can be taken.

This is a supervised binary classification problem: given a customer profile, the model must predict Churn = 1 (will leave) or Churn = 0 (will stay).

# III DataSet Description

A real Dataset provided by IBM constituting of a set of 7032 customers and their information

## Features, their types and descriptions:

customerID Text – Unique identifier for each customer
gender Binary – Male or Female
SeniorCitizen Binary – Whether the customer is a senior citizen (0 / 1)
Partner Binary – Whether the customer has a partner (Yes / No)
Dependents Binary – Whether the customer has dependents (Yes / No)
tenure Numeric – Months the customer has been with the company (1–72)
PhoneService Binary – Whether the customer has phone service (Yes / No)
MultipleLines Categorical – Single line / Multiple lines / No phone service
InternetService Categorical – DSL / Fiber optic / None
OnlineSecurity Categorical – Online security add-on (Yes / No / No internet)
OnlineBackup Categorical – Online backup add-on (Yes / No / No internet)
DeviceProtection Categorical – Device protection add-on (Yes / No / No internet)
TechSupport Categorical – Tech support add-on (Yes / No / No internet)
Streaming TVCategorical – TV streaming service (Yes / No / No internet)
StreamingMovies Categorical – Movie streaming service (Yes / No / No internet)
Contract Categorical – Month-to-Month / One Year / Two Year
PaperlessBilling Binary – Whether the customer uses paperless billing (Yes / No)
PaymentMethod Categorical – Electronic Check / Mailed Check / Bank Transfer / Credit Card
MonthlyCharges Numeric – Monthly billing amount ($18–$119)
TotalCharges Numeric – Cumulative billing amount to date
Churn Binary – Whether the customer left the company (Yes / No) (Target variable)

# VI Methodology

## VI-1 Random Forest

For a business problem like churn prediction — with nonlinear interactions between contract type, tenure, and billing — Random Forest is an excellent fit as it naturally captures these complex relationships without requiring manual feature engineering.

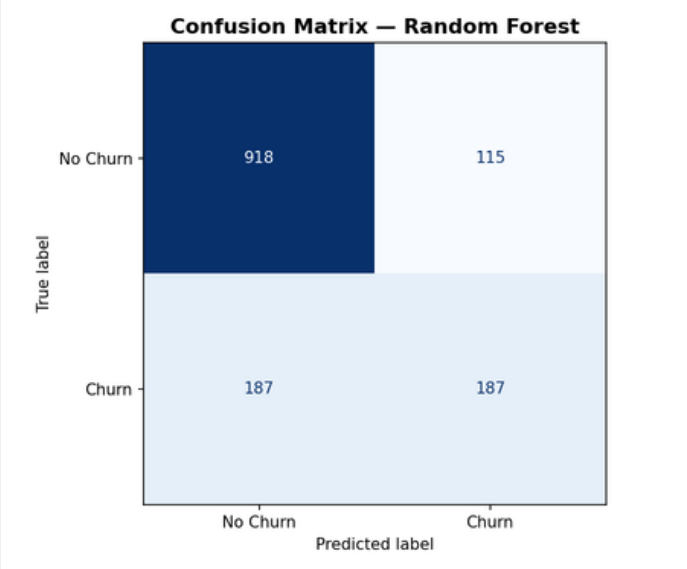
## VI-2 Data processing

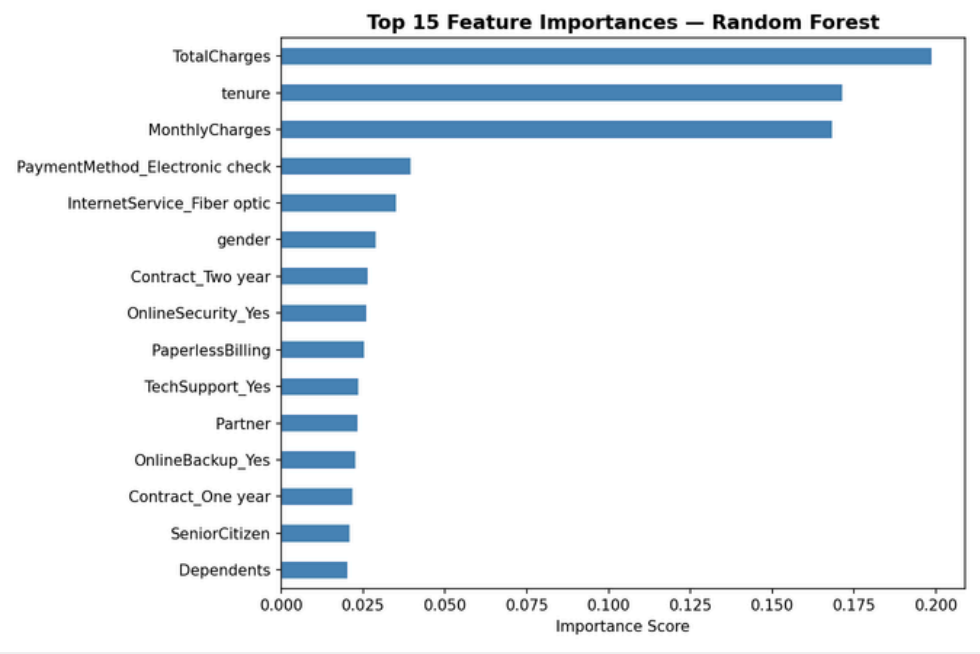
1. Drop customerID — Removed since it's just an identifier with no predictive value.
2. Fix TotalCharges — Column was stored as text instead of a number. Converted it to numeric, which revealed 11 blank rows that were then dropped.
3. Encode the target (Churn) — Converted "Yes" → 1 and "No" → 0 so the model can process it.
4. Encode binary columns — Columns like Partner, Dependents, PhoneService, and PaperlessBilling were converted from "Yes/No" text to 1/0.
5. Encode gender — Converted "Male" → 1 and "Female" → 0.
6. One-Hot Encode multi-category columns — Columns with more than 2 options (like Contract, InternetService, PaymentMethod) were expanded into separate 0/1 columns so the model doesn't assume any ranking between categories.
7. Verify no missing values — Confirmed 0 missing values remaining before training.

## VI-3 Python code

```
#    Step 1 → Load the dataset
#    Step 2 → Clean the data (fix types, remove bad rows, encode categories)
#    Step 3 → Split data into Training and Testing sets
#    Step 4 → Train a Random Forest model
#    Step 5 → Evaluate the model (accuracy, precision, recall, F1)
#    Step 6 → Show which features matter most
```

# V Model Results and valuation



- Accuracy:

78.54% The model correctly predicted whether a customer would churn or not about 4 out of every 5 customers.

- Confusion Matrix:
  - Correctly identified 918 customers who stayed
  - Correctly identified 187 customers who churned
  - Missed 187 actual churners (predicted they'd stay)
  - Wrongly flagged 115 loyal customers as churners
-
- Precision & Recall:
  - For No Churn: Precision 83%, Recall 89% – the model is very good at identifying loyal customers
  - For Churn: Precision 62%, Recall 50% – it catches only half of actual churners, meaning it struggles more with the minority class

- Feature Importance:
  - TotalCharges – customers with higher total bills are more complex to predict
  - Tenure – how long they've been with the company is a strong signal
  - MonthlyCharges – higher monthly bills correlate with higher churn risk